

Supplementary material of the morpho-chromatic analyses

Table 1. Specimens of *Chamaeza campanisona* measured and photographed in Brazilian ornithological collections, with their respective voucher codes, localities and group assignment. Acronyms: Museu Nacional, Universidade Federal do Rio de Janeiro (MN); Museu de Diversidade Biológica, Área Zoologia, Universidade Estadual de Campinas (ZUEC); Museu de Zoologia da Universidade de São Paulo (MZUSP).

Voucher	Locality	Group
MN27240	Brazil, Espírito Santo	<i>campanisona</i>
MN26554	Brazil, Espírito Santo, Jatiboca	<i>campanisona</i>
MZUSP28273	Brazil, Espírito Santo, Santa Leopoldina, Chaves	<i>campanisona</i>
MZUSP28274	Brazil, Espírito Santo, Santa Leopoldina, Chaves	<i>campanisona</i>
MZUSP28275	Brazil, Espírito Santo, Santa Leopoldina, Chaves	<i>campanisona</i>
MZUSP28276	Brazil, Espírito Santo, Santa Leopoldina, Chaves	<i>campanisona</i>
MN21072	Brazil, Espírito Santo, Santa Teresa	<i>campanisona</i>
MN35578	Brazil, Espírito Santo, Santa Teresa	<i>campanisona</i>
MN44050	Brazil, Espírito Santo, Santa Teresa, Estação Biológica Santa Lúcia	<i>campanisona</i>
MZUSP25678	Brazil, Minas Gerais, Alto Rio Doce (margem direita)	<i>campanisona</i>
MZUSP25677	Brazil, Minas Gerais, Rio Doce, baixo Suaçuí	<i>campanisona</i>
MZUSP6968	Brazil, Paraná, Fazenda Monte Alegre	<i>campanisona</i>
MZUSP1856	Brazil, Paraná, Jacarézinho, Ourinho	<i>campanisona</i>
MZUSP75661	Brazil, Paraná, Ortigueira	<i>campanisona</i>
MZUSP75662	Brazil, Paraná, Ortigueira	<i>campanisona</i>
MN38112	Brazil, Paraná, Pinhão, Ribeirão estreito, Vila UHE-Segredo	<i>campanisona</i>
MN38427	Brazil, Paraná, Piraquara, Mananciais da Serra	<i>campanisona</i>
MZUSP106345	Brazil, Paraná, Tijucas do Sul, Lagoinha, Fazenda Triângulo	<i>campanisona</i>
MN54439	Brazil, Rio de Janeiro, Miguel Pereira, Reserva Biológica do Tinguá	<i>campanisona</i>
MN33181	Brazil, Rio de Janeiro, Nova Iguaçu, Tinguá	<i>campanisona</i>
MN20922	Brazil, Rio de Janeiro, Parati, Pedra Branca	<i>campanisona</i>
MN20948	Brazil, Rio de Janeiro, Parati, Pedra Branca	<i>campanisona</i>
MN24541	Brazil, Rio de Janeiro, Teresópolis, Fazenda C. Guinle	<i>campanisona</i>
MZUSP33411	Brazil, Rio de Janeiro, Teresópolis, Fazenda C. Guinle	<i>campanisona</i>

Continuation of the Table 1

Voucher	Locality	Group
MZUSP91217	Brazil, Santa Catarina, Nova Trento	<i>campanisona</i>
MN16944	Brazil, Santa Catarina, São Bento do Sul, Fazenda Naderer	<i>campanisona</i>
MZUSP854	Brazil, São Paulo, Alto da Serra	<i>campanisona</i>
MZUSP43228	Brazil, São Paulo, Anhembi, Fazenda Barreiro Rico	<i>campanisona</i>
MZUSP51421	Brazil, São Paulo, Barro Branco, Rio Guarau	<i>campanisona</i>
MZUSP103416	Brazil, São Paulo, Br 116, Km 94	<i>campanisona</i>
MZUSP24484	Brazil, São Paulo, Cantareira	<i>campanisona</i>
MZUSP43725	Brazil, São Paulo, Caraguatatuba	<i>campanisona</i>
MZUSP78875	Brazil, São Paulo, Caucaia do Alto, Reserva Florestal do Morro Grande	<i>campanisona</i>
MZUSP80747	Brazil, São Paulo, Cotia, Reserva Florestal do Morro Grande	<i>campanisona</i>
MZUSP97953	Brazil, São Paulo, Cubatão	<i>campanisona</i>
MZUSP55074	Brazil, São Paulo, Estação Engenheiro Ferraz	<i>campanisona</i>
MZUSP55075	Brazil, São Paulo, Estação Engenheiro Ferraz	<i>campanisona</i>
MZUSP55076	Brazil, São Paulo, Estação Engenheiro Ferraz	<i>campanisona</i>
MZUSP56859	Brazil, São Paulo, Estação Engenheiro Ferraz	<i>campanisona</i>
MN16945	Brazil, São Paulo, Iguape	<i>campanisona</i>
MZUSP2832	Brazil, São Paulo, Iguape	<i>campanisona</i>
MZUSP5951	Brazil, São Paulo, Ilha de São Sebastião	<i>campanisona</i>
MZUSP31023	Brazil, São Paulo, Iporanga	<i>campanisona</i>
MZUSP31024	Brazil, São Paulo, Iporanga	<i>campanisona</i>
MZUSP23964	Brazil, São Paulo, Itapura	<i>campanisona</i>
MZUSP5006	Brazil, São Paulo, Itapura	<i>campanisona</i>
MZUSP47977	Brazil, São Paulo, Onça Parda	<i>campanisona</i>
MZUSP47978	Brazil, São Paulo, Onça Parda	<i>campanisona</i>
MZUSP49975	Brazil, São Paulo, Primeiro Morro	<i>campanisona</i>
MZUSP49976	Brazil, São Paulo, Ribeirão Fundo	<i>campanisona</i>
MZUSP49985	Brazil, São Paulo, Ribeirão Fundo	<i>campanisona</i>
ZUEC2037	Brazil, São Paulo, Ribeirão Grande, P. E. Intervalles	<i>campanisona</i>
MZUSP49978	Brazil, São Paulo, Rocha	<i>campanisona</i>
MZUSP49979	Brazil, São Paulo, Rocha	<i>campanisona</i>

Continuation of the Table 1

Voucher	Locality	Group
MZUSP43638	Brazil, São Paulo, Salesópolis, Estação Biológica de Boracéia	<i>campanisona</i>
MZUSP70596	Brazil, São Paulo, Salesópolis, Estação Biológica de Boracéia	<i>campanisona</i>
MZUSP83481	Brazil, São Paulo, Salesópolis, Estação Biológica de Boracéia	<i>campanisona</i>
MZUSP11680	Brazil, São Paulo, São Miguel Arcanjo, Taquaral	<i>campanisona</i>
MZUSP103415	Brazil, São Paulo, São Paulo	<i>campanisona</i>
MZUSP24485	Brazil, São Paulo, Serra da Cantareira	<i>campanisona</i>
MZUSP60815	Brazil, São Paulo, Serra da Cantareira	<i>campanisona</i>
MZUSP15004	Brazil, São Paulo, Tabatinguara, Cananéia	<i>campanisona</i>
MZUSP47973	Brazil, São Paulo, Tamanduá, Rio Ipiranga	<i>campanisona</i>
MZUSP47974	Brazil, São Paulo, Tamanduá, Rio Ipiranga	<i>campanisona</i>
MZUSP47976	Brazil, São Paulo, Tamanduá, Rio Ipiranga	<i>campanisona</i>
MZUSP5509	Brazil, São Paulo, Ubatuba	<i>campanisona</i>
MZUSP13624	Colombia, Buena Vista, E. Andes (Above Villavicencio)	<i>columbiana</i>
MN35746	Brazil, Alagoas, Quebrangulo, Fazenda Pedra Talhada	North SFR
MN36377	Brazil, Ceará, Guaramiranga, Monte Belo	North SFR
MN52573	Brazil, Ceará, Guaramiranga, RPPN Sítio Lagoa	North SFR
MN52574	Brazil, Ceará, Guaramiranga, RPPN Sítio Lagoa	North SFR
MN52575	Brazil, Ceará, Guaramiranga, RPPN Sítio Lagoa	North SFR
MN23534	Brazil, Ceará, Pacoti, Serra de Baturité	North SFR
MN22964	Brazil, Ceará, Pacoti, Serra de Baturité, Guaramiranga	North SFR
MZUSP33408	Brazil, Pacoti, Serra de Baturité	North SFR
ZUEC847	Brazil, Mato Grosso do Sul, Caarapó	<i>tshororo</i>
MZUSP573	Brazil, Rio Grande do Sul, Novo Hamburgo	<i>tshororo</i>
MZUSP38655	Brazil, Rio Grande do Sul, Santa Maria	<i>tshororo</i>
MZUSP38656	Brazil, Rio Grande do Sul, Santa Maria	<i>tshororo</i>
MN16949	Brazil, Rio Grande do Sul, São João do Monte Negro	<i>tshororo</i>
MN52329	Brazil, Rio Grande do Sul, Tenente Portela, Centro Novo	<i>tshororo</i>

Table 2. Type-specimens analyzed only by photographs. Acronyms: Zoologisches Museum Berlin (ZMB); American Museum of Natural History (AMNH); Muséum d'Histoire Naturelle de Paris (MNHN); Muséum d'Histoire Naturelle de Neuchâtel (MHNN); United States National Museum Smithsonian (USNM); British Museum of Natural History (BMNH); Zoologische Staatssammlung München (ZSM); Phelps Collection (PC).

Taxon	Voucher	Locality
<i>Turdus brevicaudus</i> Vieillot, 1818	MNHN 4124.307	Brazil
<i>Myiothera campanisona</i> Lichtenstein, 1823	ZMB 3163	São Paulo, Brazil
<i>Myioturdus marginatus</i> Wied, 1835	AMNH 5405, 5406	Arraial da Conquista, Brazil
<i>Chamaeza olivacea</i> Tschudi, 1844	ZMB 4733; MHNN 92.6492; USNM 41916	Peru
<i>Chamaeza fulvescens</i> Salvin & Godman, 1882	BMNH 1889.7.10.825- 827; USNM 90594	Merumé Mountains, Guyana
<i>Chamaeza brevicauda venezuelana</i> Menegaux & Hellmayr, 1906	MNHN 1856.462; 4125.461	Caracas, Venezuela
<i>Chamaeza brevicauda boliviana</i> Hellmayr & von Seilern, 1912	ZSM 1931	Yungas, Bolivia
<i>Chamaeza columbiana punctigula</i> Chapman, 1924	AMNH 179367	Rio Suno, Ecuador
<i>Chamaeza brevicauda obscura</i> Zimmer & Phelps, 1944	AMNH 324715	Bolívar, Venezuela
<i>Chamaeza brevicauda yavii</i> Phelps & Phelps, 1947	PC 37704	Cerro Yavi, Venezuela
<i>Chamaeza campanisona huachamacarii</i> Phelps & Phelps, 1951	PC 51604	Cerro Huachamacare, Venezuela

Table 3. Correlation of the morpho-chromatic variables along the principal components (PC) selected. To be used for a biological interpretation of the PCs. The suffix *_L*, *_a* and *_b* refers to the CIELab lightness (L*), green-red (a*) and blue-yellow (b*), respectively.

Variable	PC1	PC2	PC3	PC4	PC5	PC6	PC7
wing_chord	0.12	-0.27	0.54	0.28	0.25	0.33	0.12
underTailCoverts_L	0.04	-0.19	-0.39	0.02	-0.21	-0.17	0.61
underTailCoverts_b	0.19	0.19	0.54	-0.52	-0.31	0.33	0.17
underTailCoverts_a	0.12	0.23	0.57	-0.62	-0.21	0.24	0.01
tarsus_length	-0.05	0.00	0.45	0.01	0.27	0.05	0.58
tail_length	-0.02	-0.10	0.40	0.35	-0.07	0.18	-0.32
nape_L	0.59	-0.43	-0.26	-0.18	0.15	0.04	0.17
nape_b	0.59	-0.03	-0.03	-0.33	0.41	-0.19	-0.03
nape_a	0.34	0.39	0.20	-0.29	0.04	-0.51	-0.05
foreheadLaterals_L	0.35	-0.69	0.08	0.22	-0.41	-0.14	0.09
foreheadLaterals_b	0.32	-0.61	0.36	0.37	-0.38	-0.16	0.00
foreheadLaterals_a	0.48	-0.01	0.60	0.18	-0.28	-0.23	-0.10
foreheadCenter_L	0.57	0.52	-0.27	0.29	-0.32	0.12	0.17
foreheadCenter_b	0.51	0.70	-0.19	0.29	-0.25	0.14	-0.01
foreheadCenter_a	0.50	0.77	-0.13	0.22	-0.16	0.01	-0.05
crown_L	0.63	-0.37	-0.29	0.07	0.13	0.37	0.09
crown_b	0.69	-0.24	-0.14	-0.03	0.35	0.30	-0.23
crown_a	0.73	-0.02	0.03	-0.34	0.17	-0.25	-0.08
bill_width	-0.07	0.65	0.23	0.09	0.16	-0.11	0.18
bill_length	0.00	0.46	0.20	0.47	0.41	0.13	0.11
bill_height	0.19	-0.07	0.40	0.40	0.41	-0.33	0.05

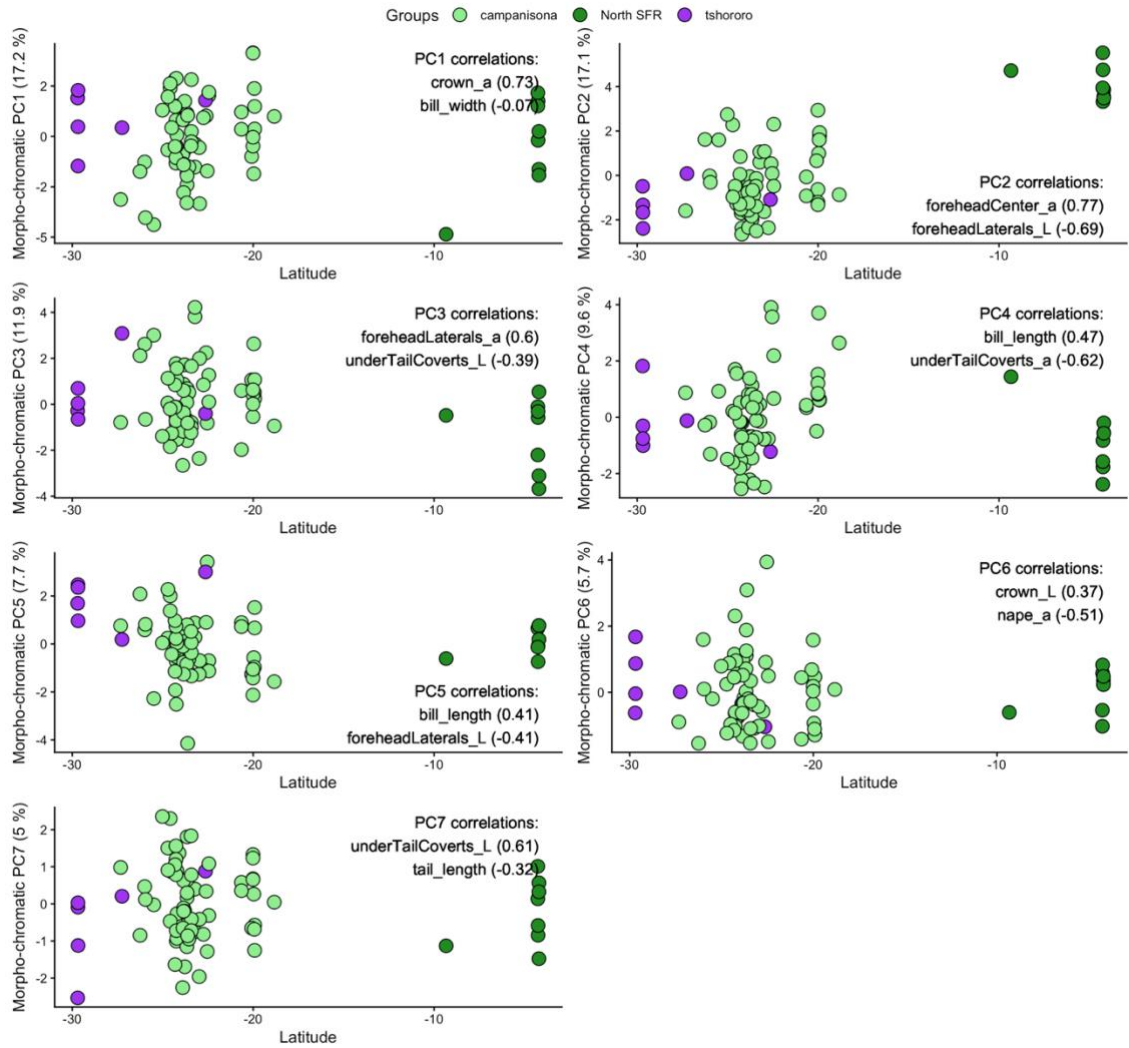


Figure 1. Variation of the principal components (PC) along the latitude, showing *C. campanisona* groups in the Atlantic Forest (*campanisona*, *tshororo* and North SFR). For a biologically interpretation of the PCs, their most strong correlation with morpho-chromatic variables is shown; positive and negative correlations may be used to a general interpretation of the latitudinal morpho-chromatic variation. See Table 3 for more correlations and a comprehensive interpretation of the PCs. The variables suffix *_L*, *_a* and *_b* refers to the CIELab lightness (*L**), green-red (*a**) and blue-yellow (*b**), respectively.